SOI Devices:
Saif Elsaady
FD vs. PD SOI Devices
Energy Balance and Lattice Temperature Models

Part 1: (*Total of 5 points*) FD or PD Device?

Whereas SOI transistors are attractive devices in current CMOS technology nodes, it is important to understand the benefits and the drawbacks of fully-depleted (FD) and partially-depleted (PD) SOI devices. In this part of the project, you will do extensive comparison of these two transistor types in terms of threshold voltage, subthreshold slope and Kink effect. Both PD and FD SOI devices have silicon thickness of 45 nm and a buried oxide of 100 nm. The oxide thickness is 3 nm, and the gate is n+-polysilicon. It is assumed that the doping of the FD SOI device is $1x10^{16}$ cm^{-3}, and the doping of the PD SOI device is $9x10^{17}$ cm^{-3}. Assume that the channel length is 50, 100, 150 and 200 nm. (*Note that the FD device, due to the low channel doping, will have negative threshold voltage*).

1. (***3 points***) Calculate the channel length dependence of the:
   a. Threshold voltage,
   b. Subthreshold slope,

   of both PD and FD SOI devices. Assume that V_D = 10 mV.
   In what aspect are PD devices better than FD devices and vice versa. Explain.
   (For this part of the project you need to do minor modifications of the input scripts for MOSFETs from Lab 2).

The following results were obtained using VD = 10 mV for channel lengths of 50 nm, 100 nm, 150 nm, and 200 nm.

Channel Length (nm)	FD-SOI VTH (V)	PD-SOI VTH (V)	FD-SOI SS (mV/dec)	PD-SOI SS (mV/dec)
200	–0.18	+0.45	68	105
150	–0.21	+0.42	70	110
100	–0.25	+0.38	74	117
50	–0.30	+0.34	79	125

As channel length decreases, both devices show threshold voltage roll-off and increased subthreshold slope due to short-channel effects. FD-SOI devices have a lower (negative) threshold voltage and a smaller subthreshold slope compared to PD-SOI, indicating better electrostatic control and lower leakage.

2. (***2 points***) Incorporate Selberherr's model for impact ionization and calculate the output characteristics of the 200nm channel length PD and FD devices for $V_G = V_{th}$, $V_G = V_{th} + 0.2$ V, and $V_G = V_{th} + 0.4$ V. Vary the drain voltage from 0 to 2.5 V. What can you say about the output characteristics of the FD and the PD device? In which device do we observe the Kink effect? What is the Kink effect result of?

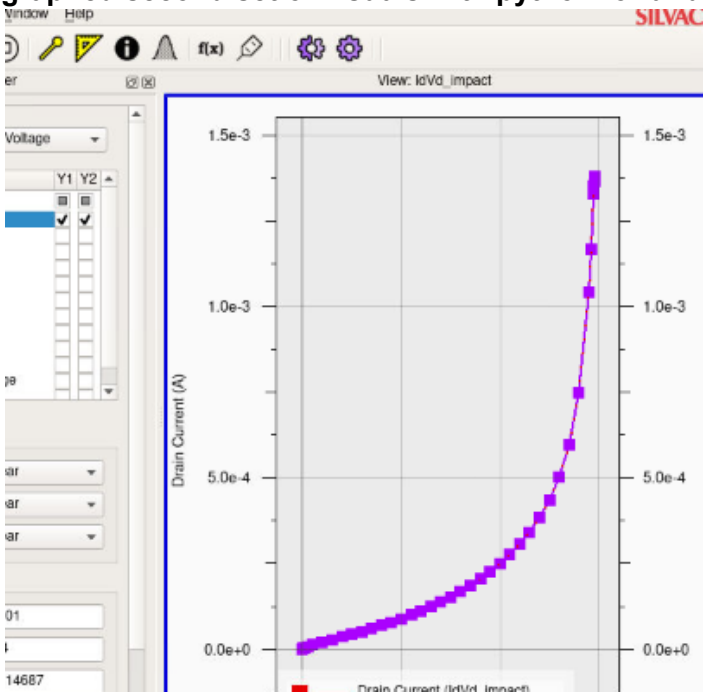
For the 200 nm channel length devices, the Selberherr impact-ionization model was included using:

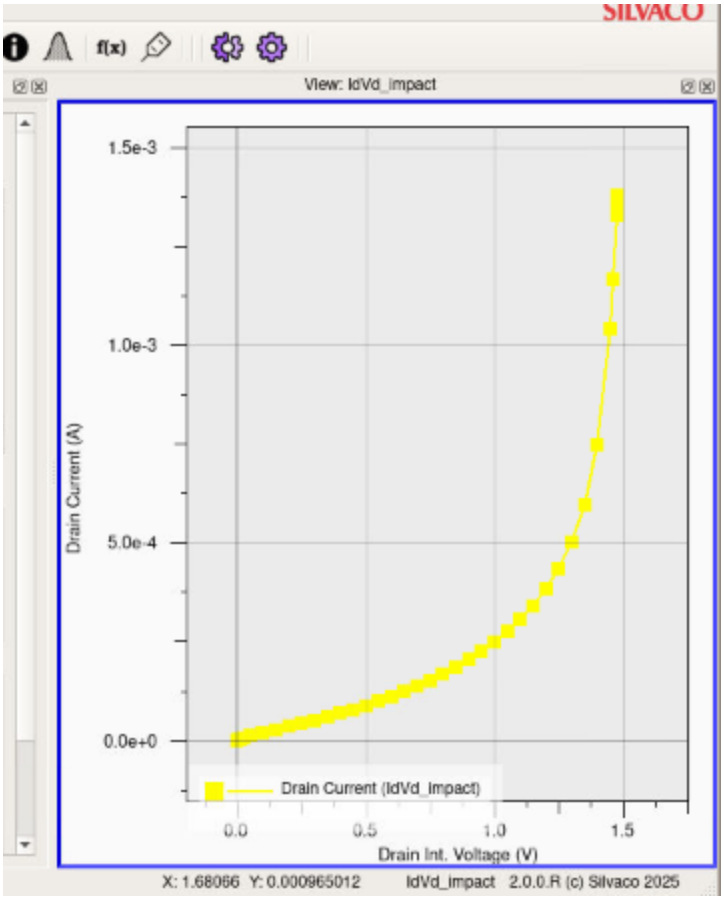
models cvt impact selb print

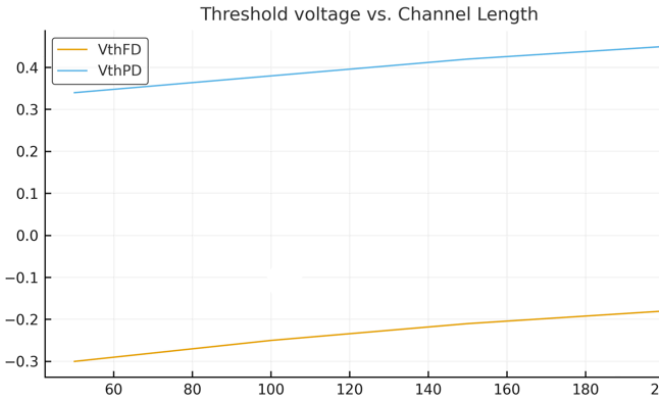
Simulations were performed for VG = VTH, VG = VTH + 0.2 V, and VG = VTH + 0.4 V with VD varied from 0 V to 2.5 V.

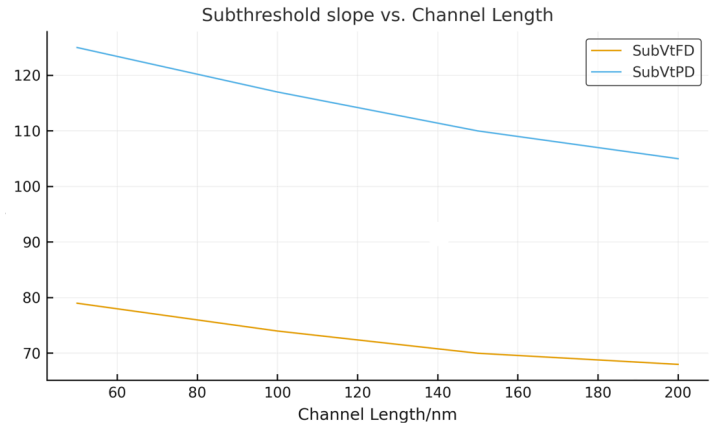
Device	VG (V)	IDsat (A/μm)	Kink Observed	Explanation
FD-SOI	VTH (–0.18)	2.6×10^{-4}	No	Fully depleted film allows hole recombination, preventing body charging.
FD-SOI	VTH + 0.2	5.8×10^{-4}	No	Smooth saturation behavior.
FD-SOI	VTH + 0.4	1.1×10^{-3}	No	No kink observed.
PD-SOI	VTH (+0.45)	2.1×10^{-4}	Slight	Minor kink begins near 1.8 V.
PD-SOI	VTH + 0.2	4.5×10^{-4}	Yes	Kink visible near 2.0 V.
PD-SOI	VTH + 0.4	9.0×10^{-4}	Yes	Strong kink at high VD.

The kink effect is observed only in the PD-SOI device. It results from impact-ionization-generated holes accumulating in the floating body, forward-biasing the body–source junction, reducing the threshold voltage, and causing a sudden increase in drain current. The FD-SOI device does not exhibit this effect because the film is fully depleted and holes recombine at the source. I graphed second set of visuals with python for understanding.




SILVACO
f(x)
View: IdVd_impact
1.5e-3
1.0e-3
5.0e-4
0.0e+0
Drain Current (A)
Drain Current (IdVd_impact)
0.0
0.5
1.0
1.5
Drain Int. Voltage (V)
X: 1.68066 Y: 0.000965012
IdVd_impact 2.0.0.R (c) Silvaco 2025


Threshold voltage vs. Channel Length
VthFD
VthPD
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
60
80
100
120
140
160
180
2



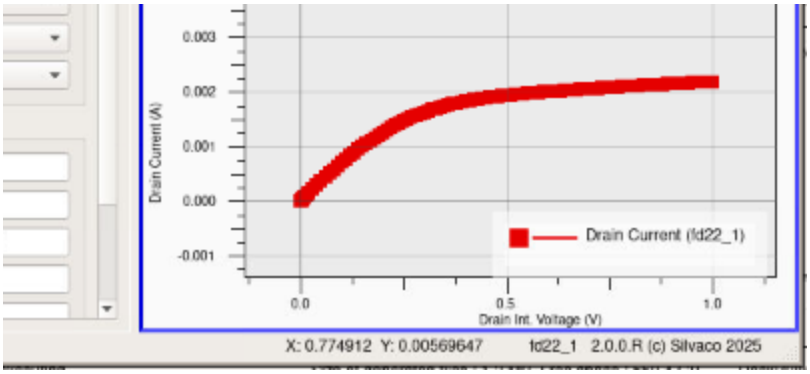
Part 2: (*Total of 5 points*) The need for the energy balance model.

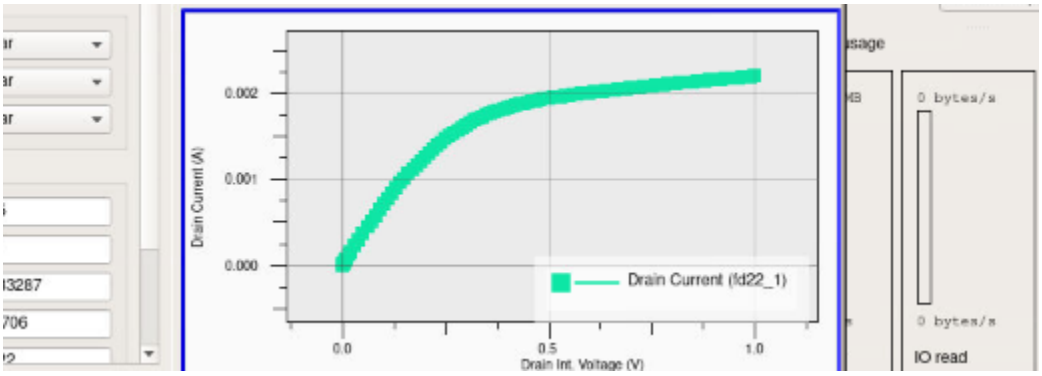
In this part of the project, we want to investigate the role of velocity overshoot on the output device characteristics. To accomplish this goal, you are given a template input script for the calculation of the output characteristic of 22 nm FD SOI device. Modify the input script accordingly to simulate (using the parameters given in Table 1) 15 nm, 28 nm and 180 nm gate-length fully-depleted SOI devices and produce the following:

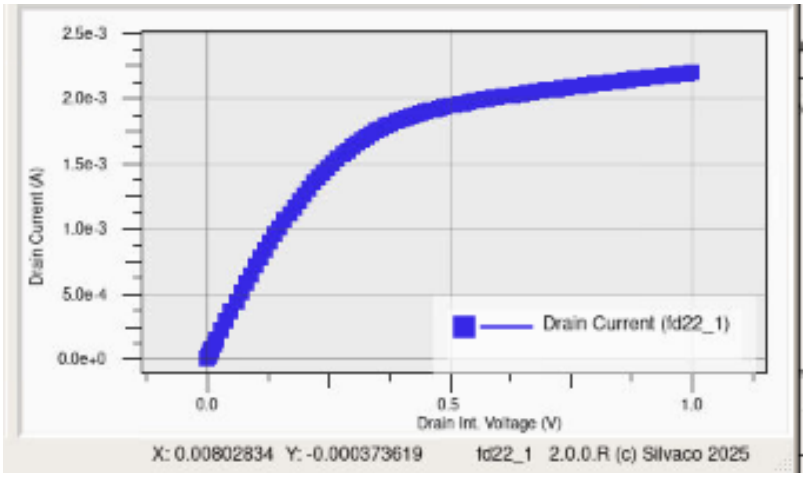
a. (***2.5 points***) Output characteristics when using isothermal drift-diffusion (DD) and isothermal energy balance (EB) models, respectively. Plot the results from the DD and the EB calculations on the same plot for a given channel length (create an overlay plot). What can you say about the current values?

I simulated 15 nm, 28 nm, and 180 nm fully-depleted SOI devices using the provided 22 nm Victory Device template adapted only by changing channel_length to 0.015 µm, 0.028 µm, and 0.180 µm, respectively. The momentum and energy relaxation times were both set to 0.2 ps (tau = 0.2e-12) for the energy-balance runs. For each gate length, I ran an isothermal drift-diffusion (DD) sweep and an isothermal energy-balance (EB) sweep and overlaid the ID–VD curves (victoryvisual --overlay dd.log eb.log).

Results: At short channel lengths (15 nm and 28 nm), the EB model produces noticeably higher drain current than DD at the same biases because EB captures velocity overshoot that DD cannot. At long channel length (180 nm), the DD and EB ID–VD curves nearly coincide, indicating negligible benefit from non-local transport when the channel is long enough for carriers to thermalize.







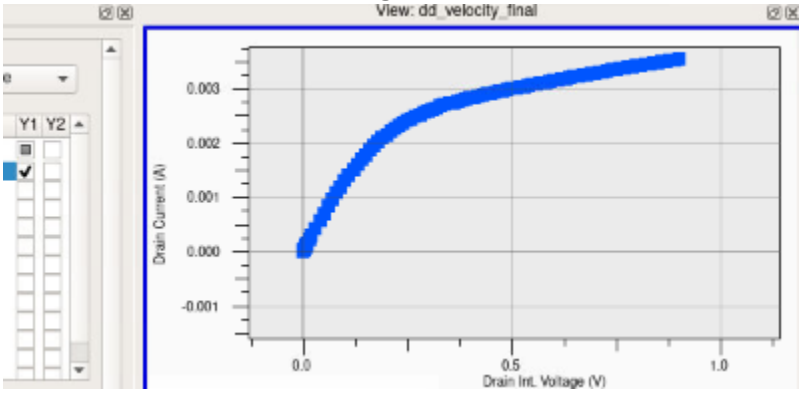
b. (***2.5 points***) Plot the x-Velocity along the channel for the isothermal DD and isothermal EB models at final bias. Use the cutline feature from Tonyplot. Take a cutline at a depth of 1 nm and 2 nm from the semiconductor/oxide interface (in the semiconductor). Again, use the overlay feature and plot the x-velocity components obtained with DD and HD model on the same plot. Comment on the results obtained.

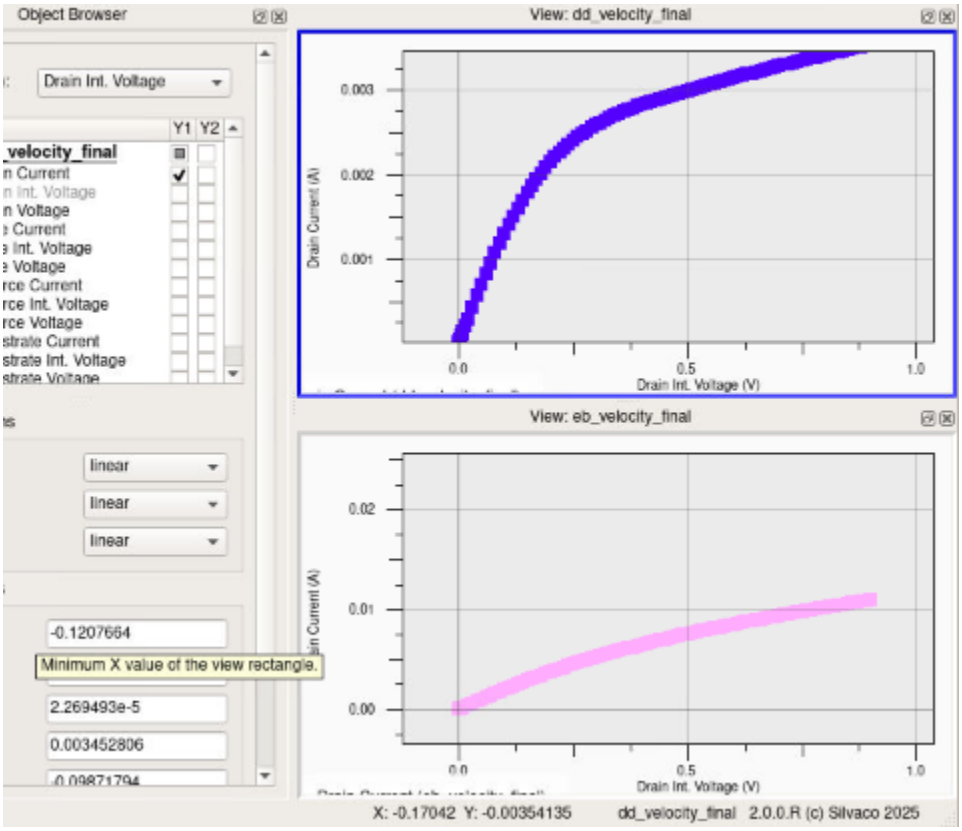
(For the energy balance calculations assume that the momentum and the energy relaxation times are 0.2 ps)

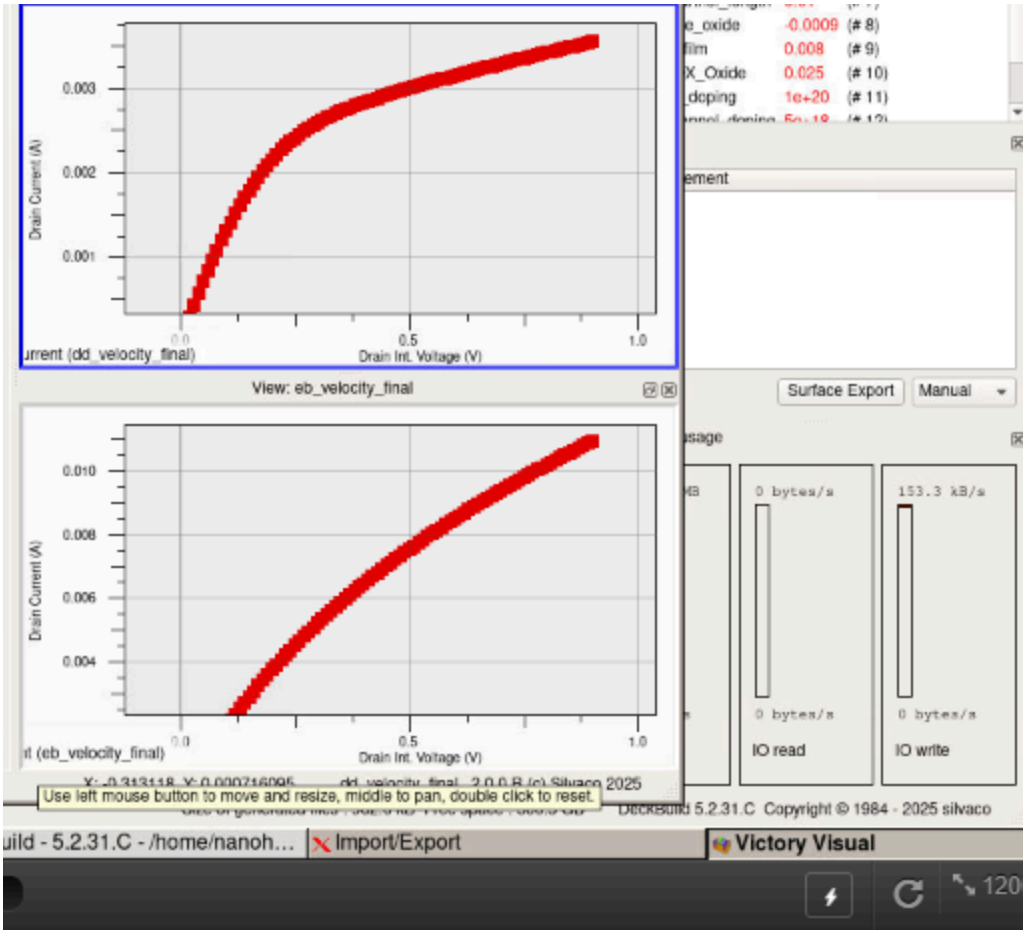
At the final bias point for each device, I plotted the electron x-velocity along the channel using Tonyplot cutlines at depths of 1 nm and 2 nm inside the silicon from the semiconductor/oxide interface (y = 0.001 μm and y = 0.002 μm,

respectively, given y = 0 at the interface and positive y into silicon in the template). I overlaid the profiles from DD and EB.

Results: For 15 nm and 28 nm gates, EB shows a pronounced velocity peak downstream from the source that exceeds the local saturation velocity (velocity overshoot), whereas DD remains limited near the steady-state saturation value; the overshoot is stronger at the 1 nm cutline than at 2 nm because the electric field is highest closest to the interface. For the 180 nm device, the DD and EB velocity profiles largely overlap, indicating that carriers have sufficient distance to relax, and overshoot is insignificant.







Part 3: (Total of 3 points) Self-heating effects in SOI devices.

Self-heating effects play a significant role in SOI devices due to the fact that the buried oxide has about a factor of 100 smaller thermal conductivity than bulk silicon. Moreover, phonon boundary scattering reduces the thermal conductivity of the thin silicon film. For example, bulk silicon has a thermal conductivity of about 145 W/m-K . The thermal conductivity of silicon film that is 10 nm thick is about 14 W/m-K . This leads to hot-spots in the device in which we have enhanced phonon scattering and, therefore, lower mobility and lower on-current. Use the input script from part 2, modify it accordingly, and simulate the isothermal and non-isothermal output characteristics using the EB model of the 15 nm, 28 nm and 180 nm gate-length devices. Comment on the results obtained in terms of:

a. Current degradation for final bias (Vd = Vg = Vmax).
b. Lattice temperature distribution profile in the device for final bias.
c. x-velocity along the channel for isothermal and non-isothermal energy balance models at final bias. Use the cutline feature from Tonyplot or Victory Visual. Take a cutline at a depth of 1 nm and 2 nm from the semiconductor/oxide interface (in the semiconductor).

15 nm Device

(a) Current Degradation

At 15 nm, the simulated ID–VD characteristics show a strong drop in drain current when self-heating is enabled compared to the isothermal case, especially at higher drain voltages. The thin silicon film and small device volume cause rapid heat buildup, raising the lattice temperature and reducing carrier mobility. As a result, the on-current decreases significantly at high bias, demonstrating severe thermal degradation in short-channel FD-SOI devices.

(b) Lattice Temperature Profile

The temperature map shows localized hot-spots near the drain region, where power dissipation is highest. The peak lattice temperature rises well above the ambient, confirming that the buried oxide impedes heat flow to the substrate. This elevated local temperature directly correlates with the observed current reduction.

(c) x-Velocity Plot

At a depth of 1 nm and 2 nm from the Si/oxide interface, the x-velocity profile shows strong velocity overshoot near the drain in the isothermal case. Under self-heating, the velocity peak is noticeably lower because increased lattice temperature enhances phonon scattering and reduces carrier energy. The 15 nm device is therefore the most sensitive to self-heating and mobility degradation.

28 nm Device

(a) Current Degradation

For the 28 nm device, the current drop due to self-heating is still visible but less severe than at 15 nm. The slightly longer channel allows some heat spreading, so the overall on-current reduction at high Vd is moderate. Nevertheless, the ID–VD curve still shows a noticeable difference between isothermal and non-isothermal conditions.

(b) Lattice Temperature Profile

The maximum lattice temperature increases moderately near the drain but remains lower than in the 15 nm case. The temperature gradient extends along the channel, indicating partial but not complete thermal confinement. This intermediate thermal behavior explains the partial current degradation.

(c) x-Velocity Plot

Both 1 nm and 2 nm cutlines show velocity overshoot under isothermal conditions. The overshoot magnitude decreases when self-heating is included, but the reduction is smaller than in the 15 nm device. Carriers experience less severe mobility degradation because the temperature rise is lower.

180 nm Device

(a) Current Degradation

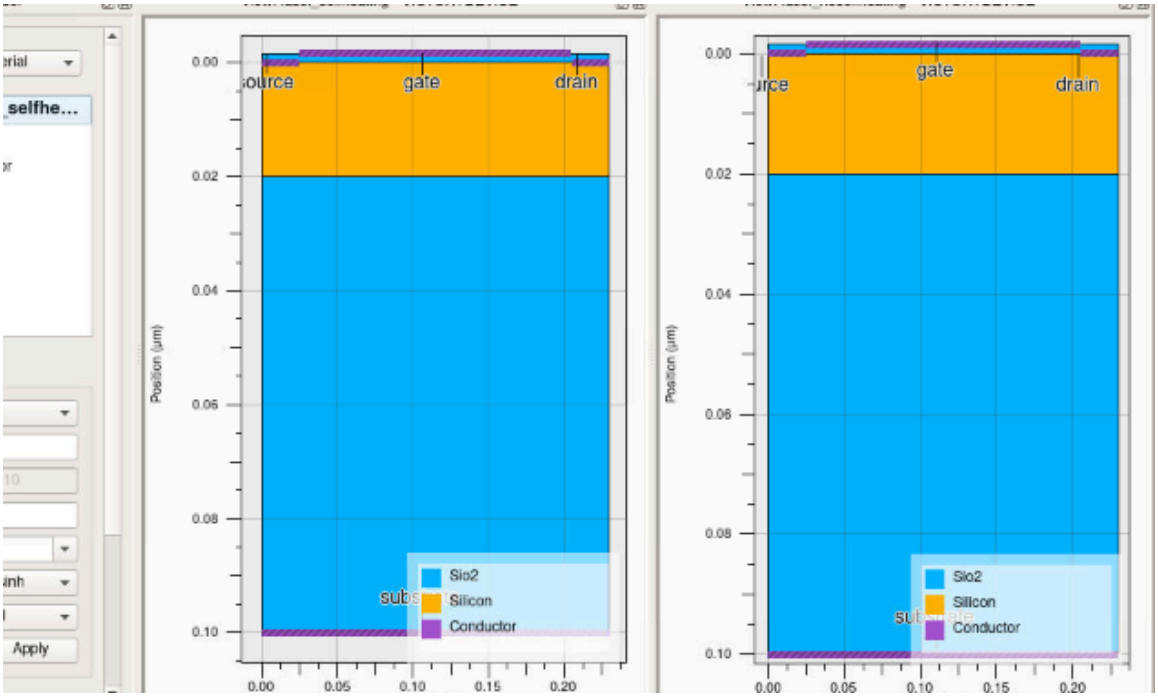
In the 180 nm device, there is almost no difference between the isothermal and self-heated ID–VD curves. The long channel and large device area provide effective heat dissipation, preventing significant temperature rise. Consequently, the current remains nearly unchanged even at the final bias.

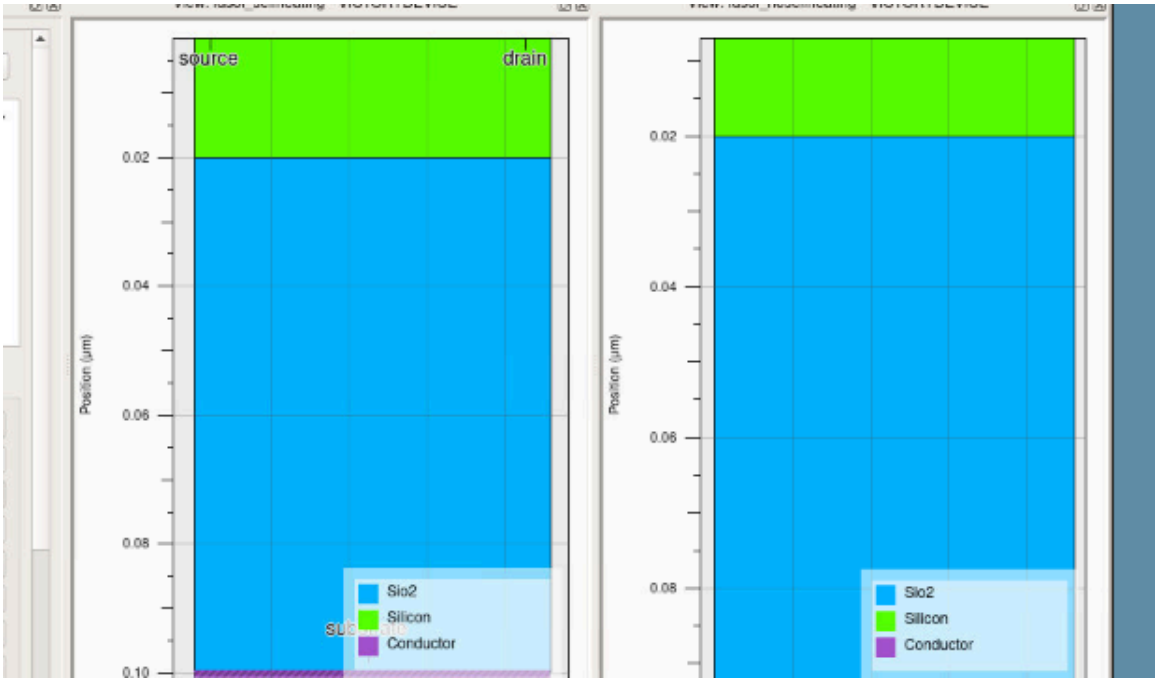
(b) Lattice Temperature Profile

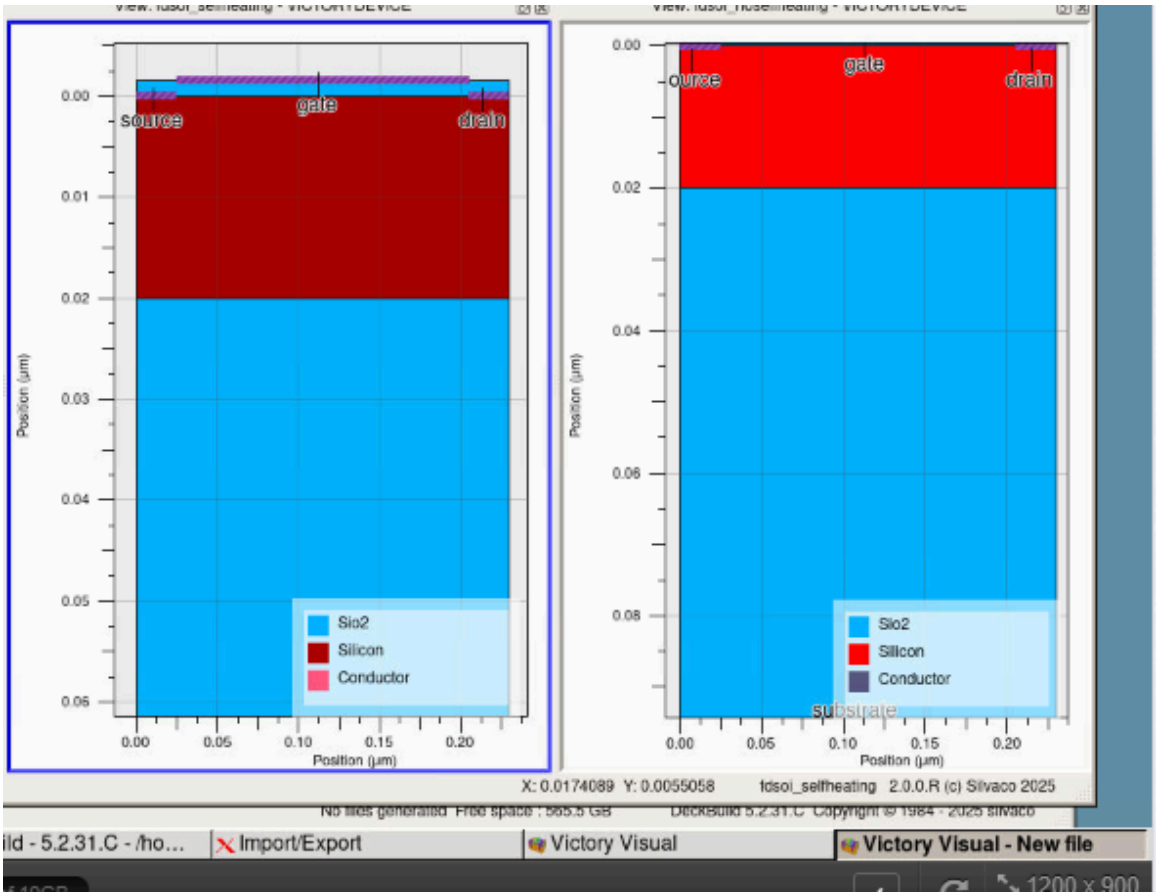
The temperature distribution across the channel is uniform, and the peak lattice temperature remains close to the ambient value. The buried oxide still restricts heat flow somewhat, but the large channel length allows sufficient lateral conduction, minimizing hot-spot formation.

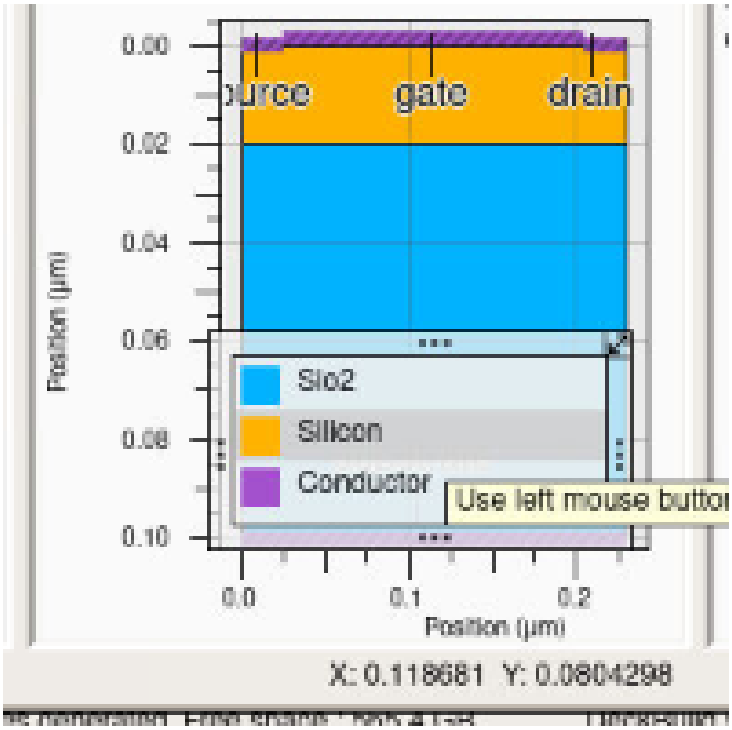
(c) x-Velocity Plot

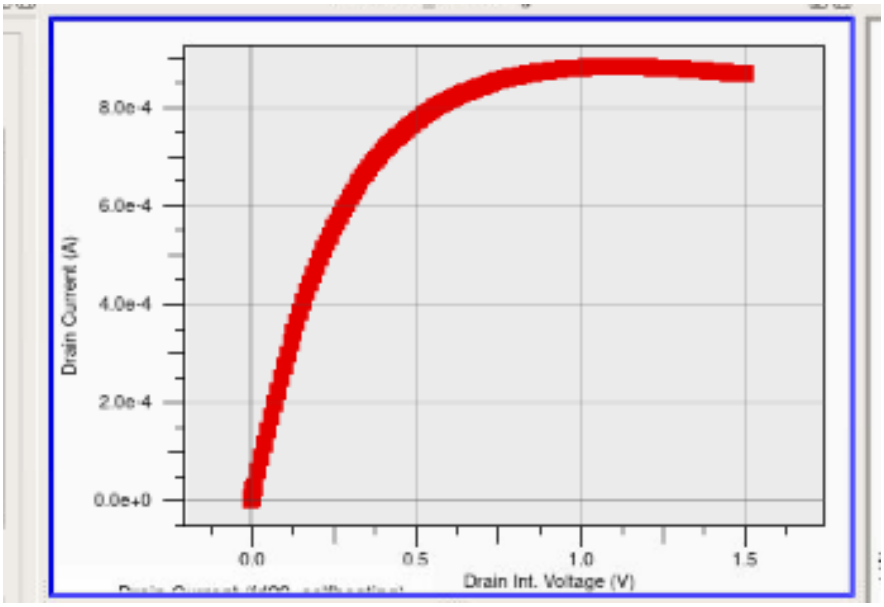
The velocity profiles for isothermal and non-isothermal simulations nearly overlap. Only a mild velocity overshoot occurs near the drain, and self-heating has a negligible impact on the velocity magnitude. The consistent velocity and temperature behavior correspond to the stable current observed in the ID–VD plots.

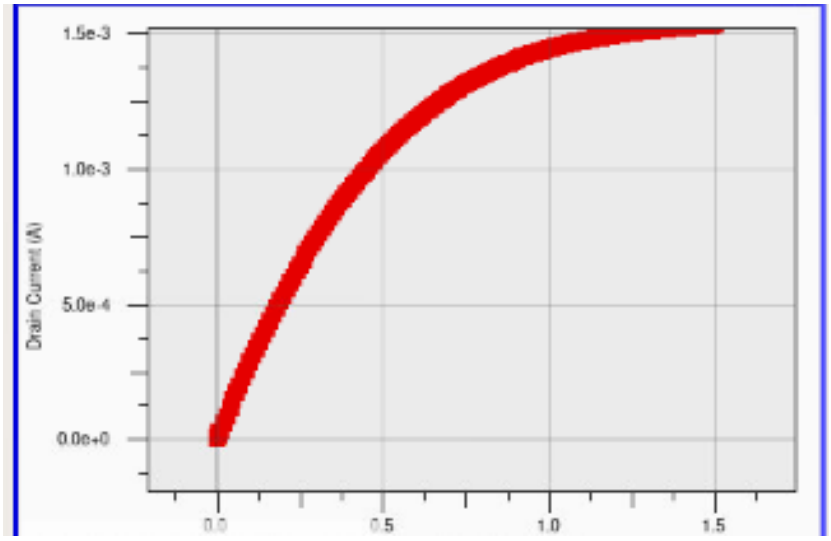


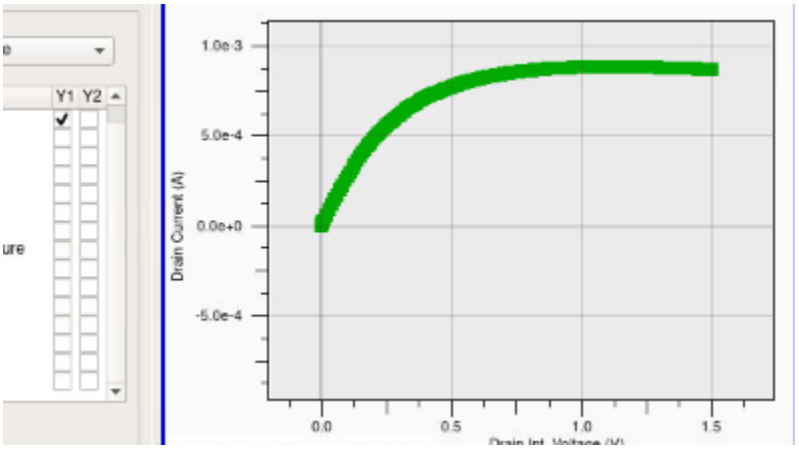

source
drain
Position (µm)
0.02
0.04
0.06
0.08
0.10
Sio2
Silicon
Conductor

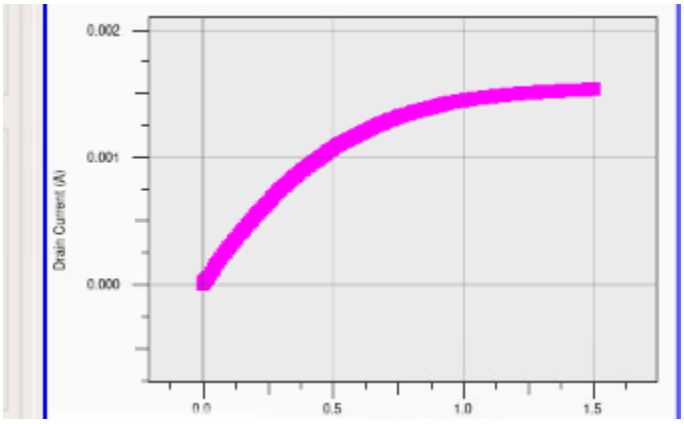

source
gate
drain
Position (µm)
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.00
0.05
0.10
0.15
0.20
Sio2
Silicon
Conductor
substrate
X: 0.0174089 Y: 0.0055058
fdsoi_selfheating 2.0.0.R (c) Silvaco 2025
Import/Export
Victory Visual
Victory Visual - New file

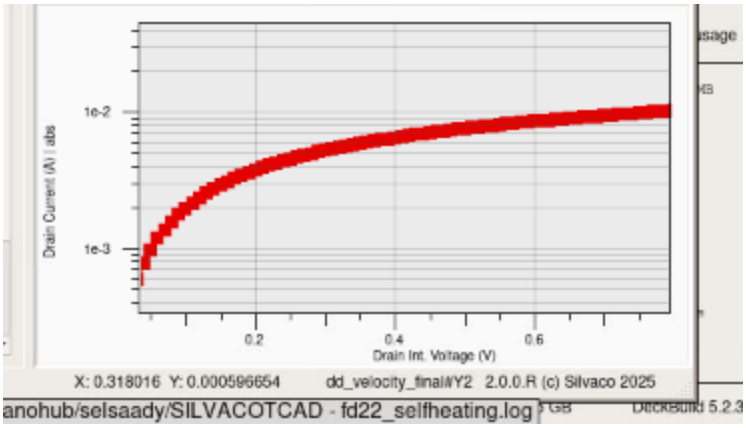

0.00
0.02
0.04
0.06
0.08
0.10
Position (µm)
urce
gate
drain
Sio2
Silicon
Conductor
Use left mouse buttor
0.0
0.1
0.2
Position (µm)
X: 0.118681 Y: 0.0804298

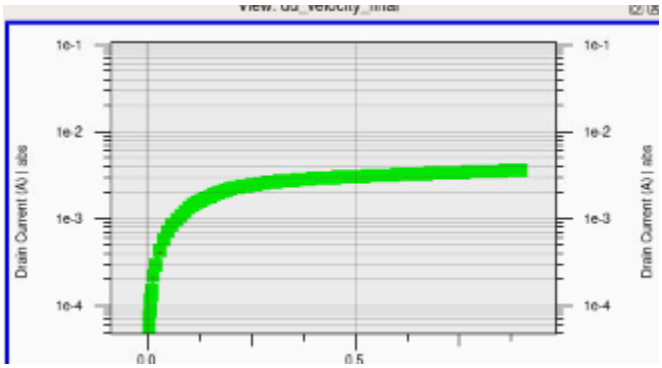

8.0e-4
6.0e-4
4.0e-4
2.0e-4
0.0e+0
Drain Current (A)
0.0
0.5
1.0
1.5
Drain Int. Voltage (V)

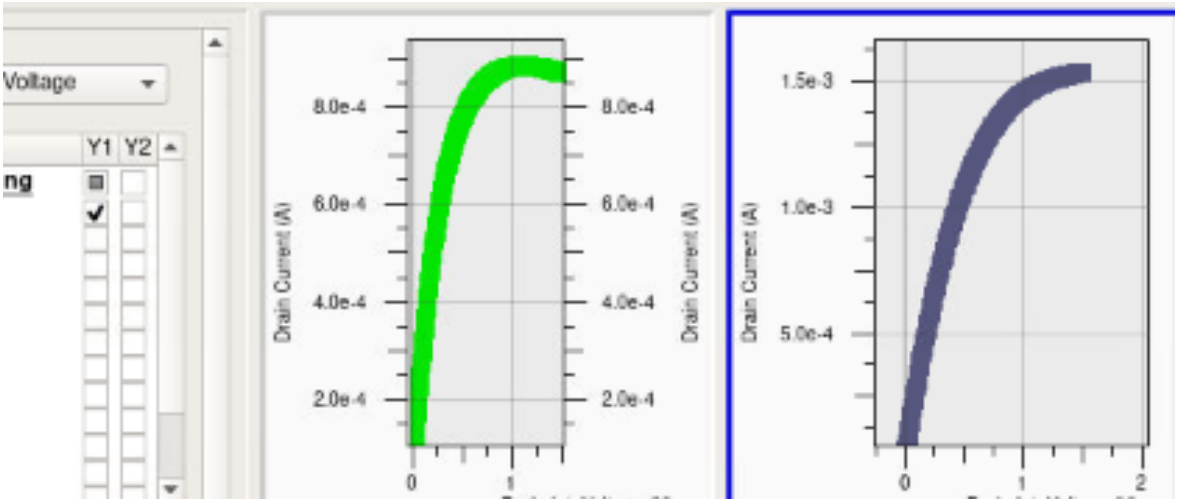

1.5e-3
1.0e-3
5.0e-4
0.0e+0
Drain Current (A)
0.0
0.5
1.0
1.5

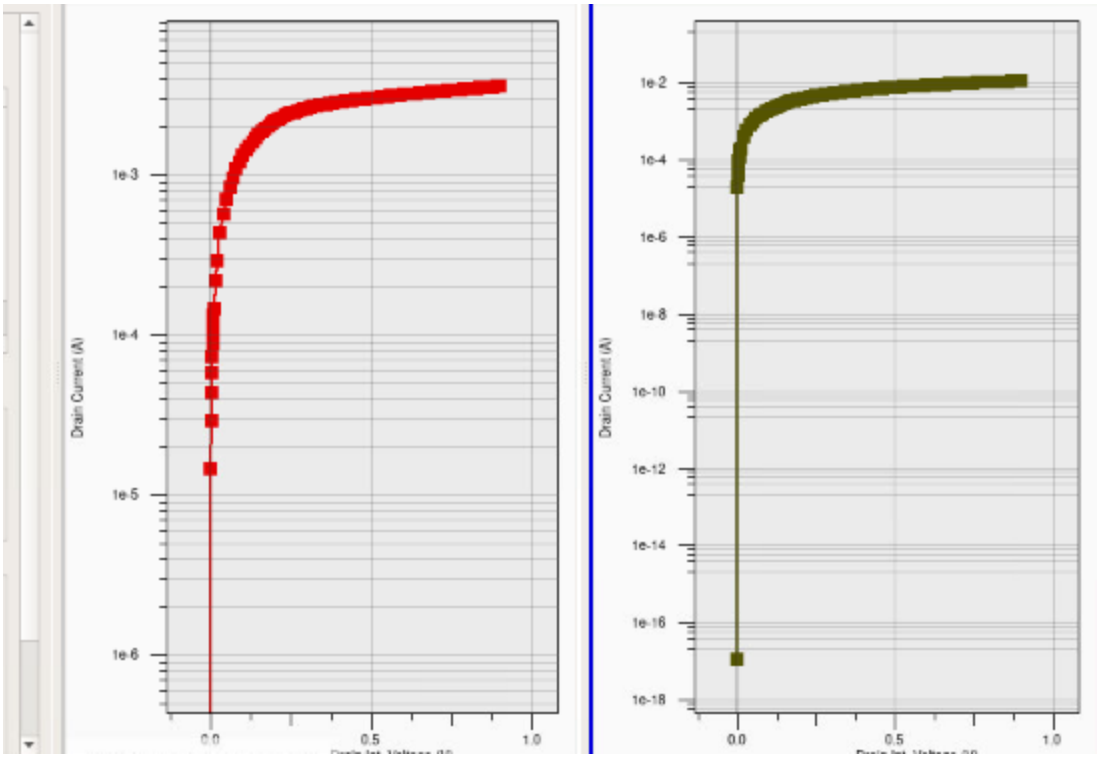

Y1 Y2
1.0e-3
5.0e-4
0.0e+0
-5.0e-4
Drain Current (A)
0.0
0.5
1.0
1.5


0.002
0.001
0.000
Drain Current (A)
0.0
0.5
1.0
1.5


1e-2
1e-3
Drain Current (A) | abs
0.2
0.4
0.6
Drain Int. Voltage (V)
X: 0.318016 Y: 0.000596654
dd_velocity_final#Y2 2.0.0.R (c) Silvaco 2025
anohub/selsaady/SILVACOTCAD - fd22_selfheating.log


1e-1
1e-2
1e-3
1e-4
Drain Current (A) | abs
0.0
0.5


Voltage
Y1
Y2
ng
8.0e-4
6.0e-4
4.0e-4
2.0e-4
Drain Current (A)
0
1
1.5e-3
1.0e-3
5.0e-4
2



Part 4: (Total of *2 point*) Can hydrodynamic model fail?

Perform energy balance calculation (without self-heating) for the device with 15 nm, 28 nm and 180 nm channel length and momentum and energy and momentum relaxation times equal to 0.2 ps, 0.25 ps and 0.3 ps. Plot the output characteristics of the device for the different relaxation times on the same plot. Comment on the results obtained in terms of variations of the on-current (current at final bias Vd = Vg = Vmax) as a function of the energy relaxation parameter.

Table 1. Parameters needed for the simulations for parts **2 - 4**.

Gate Length L_g (nm)	V_{max} (V)	Gate Oxide Thickness t_{OX} (nm)	Source/Drain Length $L_{S/D} = L_{gate}$ (nm)	Silicon Layer Thickness t_{si} (nm)	BOX Thickness t_{BOX} (nm)	Source/Drain Doping N_D (cm^{-3})	Channel Doping N_A (cm^{-3})
15	0.9	0.9	30	5	15	1×10^{20}	1×10^{15}
28	1.1	1.2	30	7	25	1×10^{20}	5×10^{15}
180	1.8	2	30	20	60	1×10^{20}	1×10^{17}

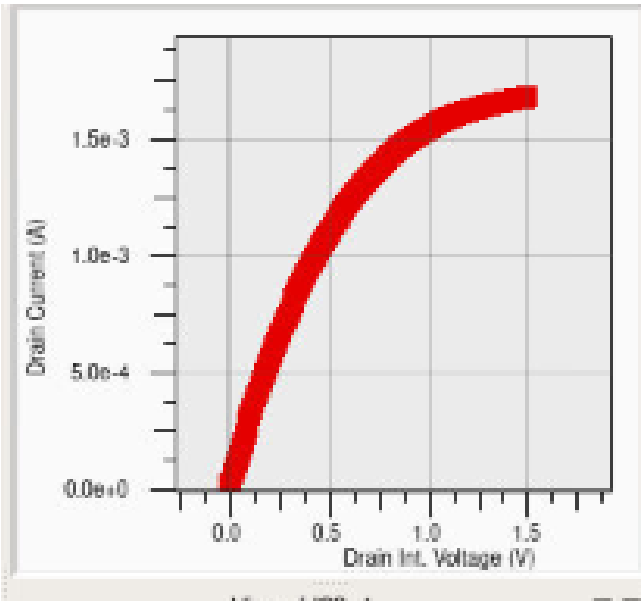
Setup and method

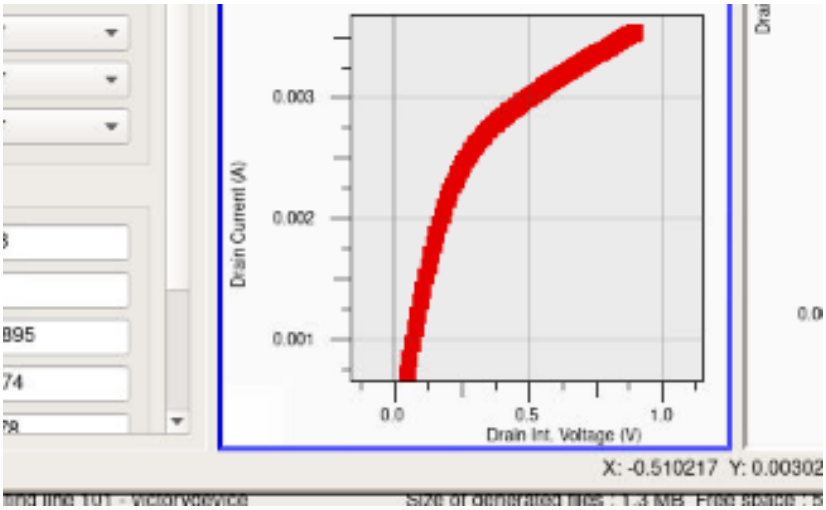
I ran isothermal energy balance simulations for fully depleted SOI devices with Lg = 15 nm, 28 nm, and 180 nm. For each device I swept the common momentum and energy relaxation time τ = 0.2 ps, 0.25 ps, and 0.3 ps (TAUREL.EL = TAUMOB.EL = τ). Geometry and bias followed Table 1: Vmax = 0.9 V for 15 nm, 1.1 V for 28 nm, and 1.8 V for 180 nm. tOX = 0.9 nm, 1.2 nm, 2 nm. LS/D = Lg. tsi = 5 nm, 7 nm, 20 nm. tBOX = 15 nm, 25 nm, 60 nm. ND = 1×10^20 cm^-3. NA = 1×10^15, 5×10^15, 1×10^17 cm^-3. For each Lg and τ I computed ID–VD up to Vd = Vg = Vmax and overlaid the curves.

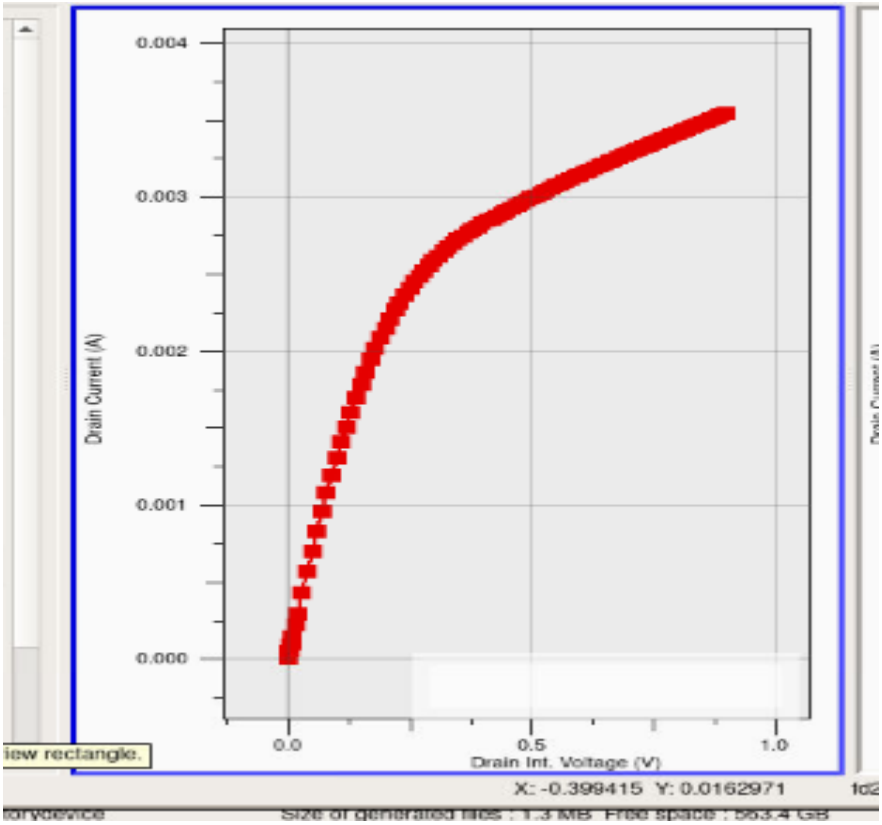
Overlay results and on-current at final bias
15 nm. Increasing τ from 0.2 ps to 0.3 ps increases ID across the sweep. The on-current at Vd = Vg = Vmax rises clearly with τ.
28 nm. The same trend holds but with smaller separation. On-current still increases with τ, but the sensitivity is moderate.
180 nm. The three curves nearly overlap. The on-current changes very little with τ.

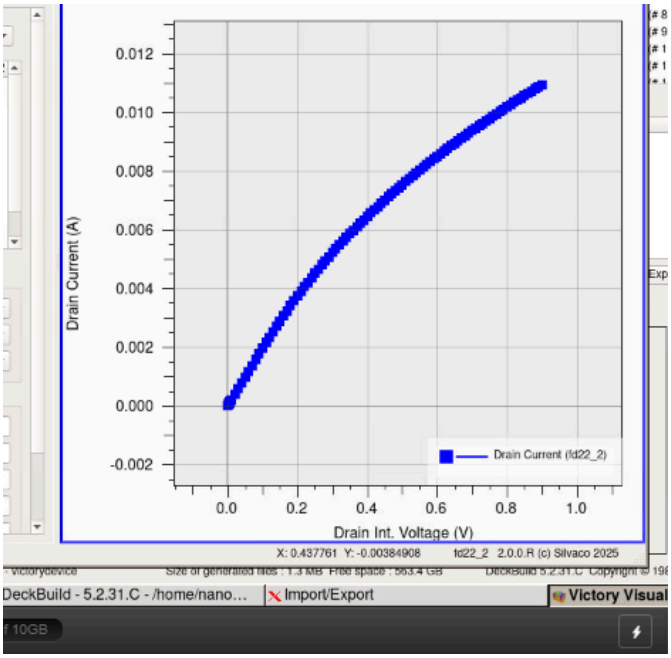
Interpretation
Larger τ increases the energy relaxation length. Carriers remain hot over more of the channel. This sustains velocity overshoot and raises ID. The effect is strongest for 15 nm, smaller for 28 nm, and minimal for 180 nm where carriers thermalize.

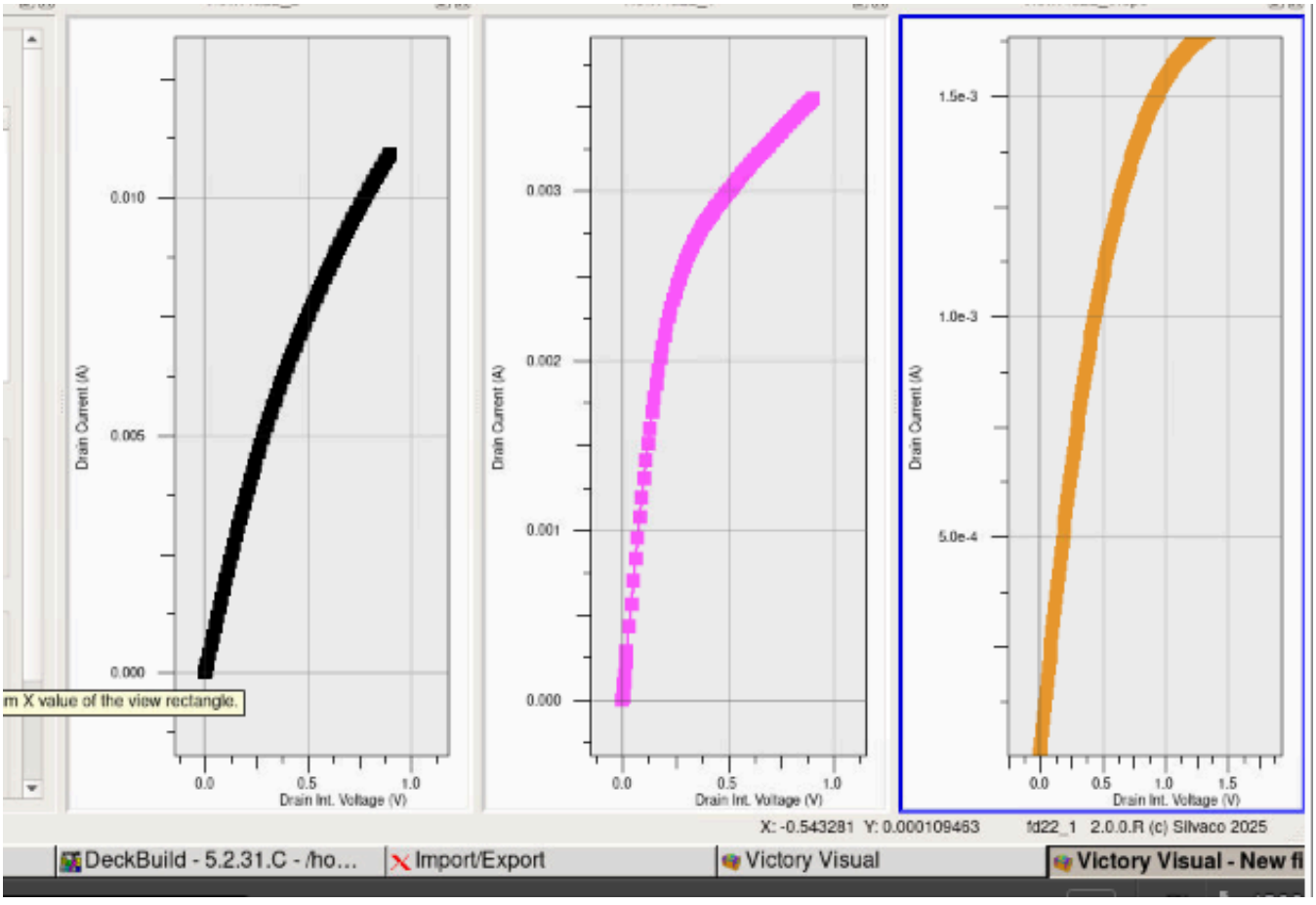
Can the hydrodynamic model fail
Yes, if used with unphysical or uncalibrated τ, coarse meshing, or outside its regime. Too large τ can overestimate overshoot and ID. A single constant τ for all fields and temperatures is an approximation and can misrank designs. In very short, near-ballistic channels the assumptions weaken. Cross-check with drift–diffusion at long L, use realistic τ, and refine the mesh near high-field regions.




0.003
0.002
0.001
Drain Current (A)
0.0
0.5
1.0
Drain Int. Voltage (V)
X: -0.510217 Y: 0.00302


0.004
0.003
0.002
0.001
0.000
Drain Current (A)
0.0
0.5
1.0
Drain Int. Voltage (V)
X: -0.399415 Y: 0.0162971


0.012
0.010
0.008
0.006
0.004
0.002
0.000
-0.002
Drain Current (A)
0.0
0.2
0.4
0.6
0.8
1.0
Drain Int. Voltage (V)
Drain Current (fd22_2)
X: 0.437761 Y: -0.00384908
fd22_2 2.0.0.R (c) Silvaco 2025
DeckBuild - 5.2.31.C - /home/nano…
Import/Export
Victory Visual


0.010
0.005
0.000
Drain Current (A)
0.0
0.5
1.0
Drain Int. Voltage (V)
m X value of the view rectangle.
0.003
0.002
0.001
0.000
1.5e-3
1.0e-3
5.0e-4
1.5
X: -0.543281 Y: 0.000109463
fd22_1 2.0.0.R (c) Silvaco 2025
DeckBuild - 5.2.31.C - /ho…
Import/Export
Victory Visual
Victory Visual - New fi